

Assignment 1 — Solutions

FinTech 545 — Quantitative Risk Management

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1 Reading the Shape of a Sample

1.1 The Moments

Table 1: The first four moments of `problem1.csv`

Moment	Value
Mean	0.001104
Standard deviation	0.009811
Skewness	-0.6688
Excess kurtosis	2.3398

At $n = 1000$ the population and sample conventions agree to four decimal places, so the choice between dividing by n and by $n - 1$ does not change anything we conclude below.

1.2 Part (a)

Table 1 rules out two of the families from Week 1, for two different reasons.

The Normal is ruled out by the excess kurtosis. A Normal has excess kurtosis 0 and the sample shows 2.34. The standard error of the excess kurtosis estimate at $n = 1000$ is roughly $\sqrt{24/n} \approx 0.155$, so 2.34 sits about fifteen standard errors away from zero. No reading of sampling noise recovers a Normal.

The symmetric Student's t is ruled out by the skewness rather than the kurtosis. A t accommodates excess kurtosis of 2.34 without difficulty, at roughly $\nu = 7$. What a t cannot produce is a skewness of -0.67 , because the t is symmetric about its mean and its third moment is zero by construction.

What survives is a family that carries skewness and excess kurtosis as separate parameters. The Normal Inverse Gaussian is the natural candidate from Week 1, and the skewed variants qualify on the same grounds. Both moments are finite and stable across subsamples, so the undefined moment cases from Week 1 do not apply here.

1.3 Part (b)

Fitting a Normal by matching the first two moments gives $\mu = 0.001104$ and $\sigma = 0.009811$. Its 1% quantile is -0.02172 . The empirical 1% quantile is -0.02699 , which is 24% further out.

26 observations fall below the Normal's 1% quantile. A Normal predicts **10**. The sample produces two and a half times as many.

1.4 Part (c)

The Normal is wrong in the left tail and close to right through the middle. At the 5% level the two nearly agree, with an empirical quantile of -0.01495 against the Normal's -0.01503 . The error is concentrated where the losses are.

The direction is that the Normal is too optimistic. It places the 1% point closer to the mean than the data does, so outcomes worse than that point arrive two and a half times as often as the model allows. Anything we compute from the far left of this fitted Normal comes out too small.

2 A Regression Whose Errors Are Not Normal

2.1 Part (a)

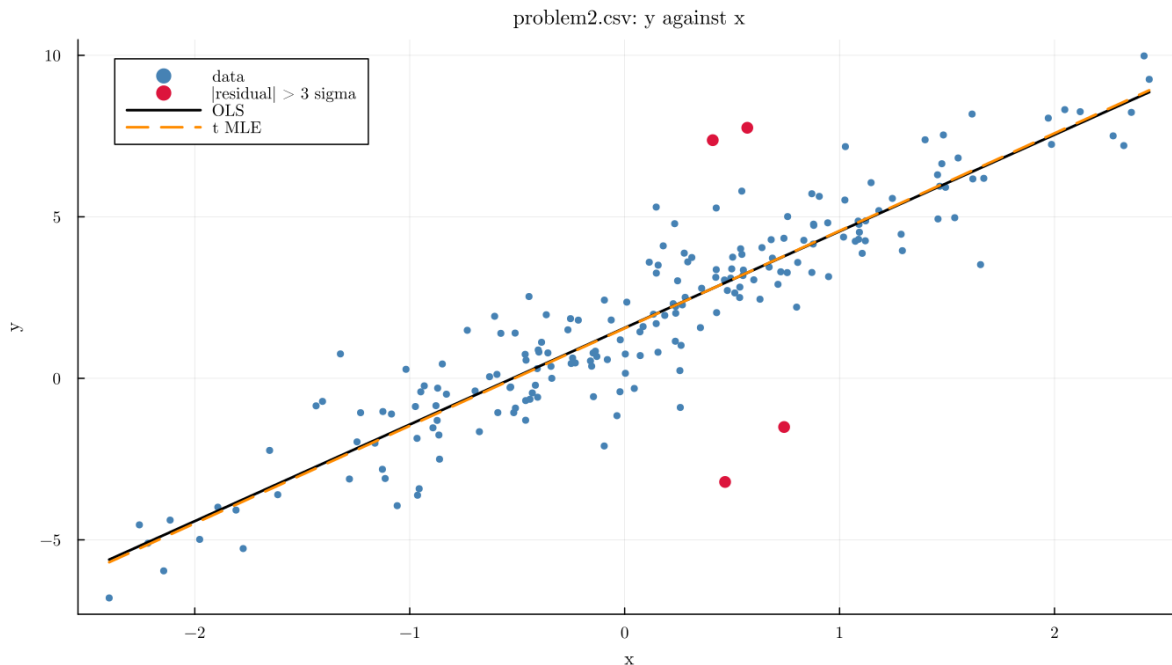


Figure 1: `problem2.csv`. Four points sit past three residual standard deviations, two above the line and two below. The OLS and t MLE lines are nearly indistinguishable

Figure 1 shows a clear linear relationship with a handful of points far off the line in both directions. Four of the 200 observations sit past three residual standard deviations, where a Normal would place well under one.

The outliers are two sided, which is the detail that decides the answer. A one sided cluster would point at a skewed error or a missing variable. A symmetric scatter of large residuals points instead at a heavy tailed symmetric error, and the Student's t is the natural candidate.

2.2 Part (b)

The violated assumption is normality of the errors, assumption 7 of the seven in Week 2.

On the slope. OLS stays unbiased under heavy tails, because unbiasedness requires only a zero mean error and not a normal one. Unbiased describes the average over repeated samples rather than the sample in hand, so a single large outlier can still move the estimate in any particular draw. We expect the outliers to pull the line, and we do not expect OLS to be biased.

On the standard error. OLS is no longer efficient. It remains unbiased but stops being best, so the estimate carries a wider sampling distribution. The usual standard error formula is unreliable, because it rests on the normal likelihood.

2.3 The Fits

Table 2: Three fits to `problem2.csv`

Model	α	β	Error scale	ν	AICc
OLS	1.5607	2.9908	1.3514	–	–
Normal MLE	1.5607	2.9908	1.3446	–	692.14
t MLE	1.5480	3.0192	0.9355	3.63	665.33

In Table 2, OLS and the Normal MLE return identical coefficients, as they must. Minimizing the sum of squares and maximizing the normal likelihood are the same problem. They differ only in the scale estimate, where OLS divides by $n - 2$ and the MLE by n .

$\Delta\text{AICc} = 26.82$ in favour of the t .

2.4 Part (c)

The three slopes are 2.9908, 2.9908, and 3.0192. In this sample the outliers moved the slope by less than one OLS standard error, because they fall on both sides and largely cancel. Figure 1 shows the mechanism: the two high points and the two low points sit at similar values of x , so their pull on the line offsets. That is also why the OLS and t lines are hard to tell apart.

This is a fact about this draw rather than a general result. Had the largest residuals landed on one side, the OLS line would have followed them. The general statement is narrower – over repeated samples the OLS slope averages to the truth, and the price of the heavy tails is paid in the spread of that average rather than in its centre.

2.5 Part (d)

What the Normal gets wrong is the shape of the error, and the visible symptom is the scale. The Normal reports 1.3446 against the t 's 0.9355, so it is 44% wider. The Normal carries one parameter to describe the spread, so it must stretch that single parameter far enough to reach the outliers. Every ordinary observation is then treated as more surprising than it was.

The t separates the two jobs. Its scale of 0.9355 describes the bulk, and $\nu = 3.63$ carries the tail.

2.6 Part (e)

Table 3: Where the two error models cross over

Quantile of the error	Normal	t
95.0%	2.212	2.054
99.0%	3.128	3.710
99.5%	3.464	4.620

Table 3 shows the Normal wider at 95% and the t wider at 99.5%. The Normal, forced to inflate its single scale parameter, overstates ordinary moves and still understates extreme ones. The crossover sits between the 95th and 99th percentiles.

For a capital buffer we want the t . The buffer is sized on the far tail, and that is precisely where the Normal is thin. At the 99.5% point the Normal gives 3.46 against the t 's 4.62, understating the loss by roughly a third. The Normal looks more conservative at 95%, but that is the wrong row to read for a buffer.

3 Pearson Against Spearman

3.1 Part (a)

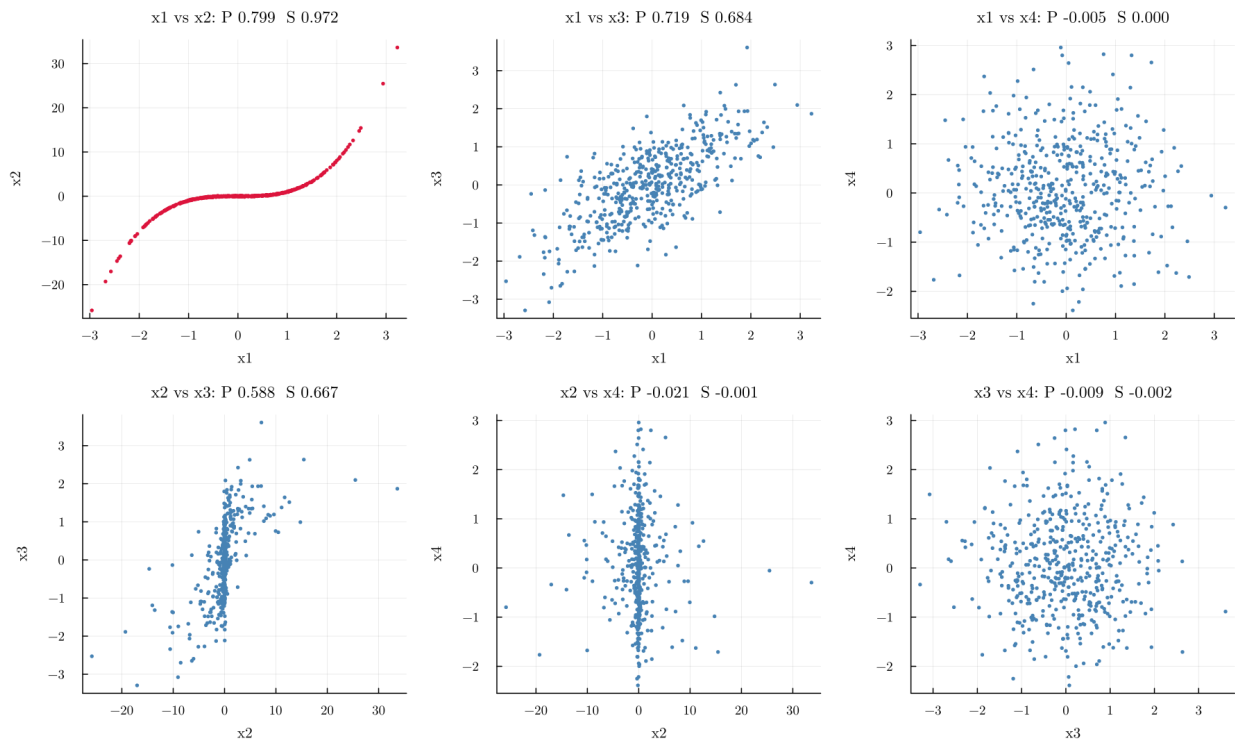


Figure 2: All six pairs from `problem3.csv`. The $x1$ against $x2$ panel, in red, is the only one that is monotone without being linear

Figure 2 settles the prediction before any coefficient is computed. Four series give six pairs, and only one of them departs from a straight line.

$x1$ against $x2$ is a curve rather than a line, tight and strictly increasing, so we expect Pearson and Spearman to disagree there and by a wide margin. $x1$ against $x3$ is a cloud around a line, so we expect the two measures to agree. $x2$ against $x3$ inherits the curvature through $x1$ and should show a smaller version of the same gap.

The three pairs involving x_4 are structureless clouds. Both coefficients should sit near zero, and they should agree with each other, since there is nothing for either to find.

3.2 Parts (b) and (c)

Table 4: Pearson against Spearman, sorted by gap

Pair	Pearson	Spearman	Gap
x1, x2	0.799	0.972	0.173
x2, x3	0.588	0.667	0.080
x1, x3	0.719	0.684	0.036
x2, x4	-0.021	-0.001	0.020
x3, x4	-0.009	-0.002	0.006
x1, x4	-0.005	0.000	0.005

In Table 4 the x_1, x_2 pair carries by far the largest gap, at 0.173. The relationship is monotone but not linear. Pearson measures how close the pair sits to a straight line, and a cubic is not a straight line, so Pearson reports 0.799. Spearman measures whether the pair moves in the same direction, and the cubic is strictly increasing, so Spearman reports 0.972 – close to the 1.0 that a noiseless monotone transform would give.

Neither coefficient is wrong. They answer different questions. Spearman is the honest description of the dependence here, because the two series move together without exception. Pearson is answering how useful a linear model would be, and 0.799 is the correct answer to that question.

The x_1, x_3 pair is the control. It is linear, and the two coefficients agree to within 0.04. The pairs involving x_4 sit near zero under both measures, as Figure 2 leads us to expect.

4 Conditional Distributions

The realized correlation between x_1 and x_2 is 0.7986.

4.1 Parts (a) and (b)

Week 2 gives the conditional distribution of a partitioned normal vector. For $X \sim N(\mu, \Sigma)$ split into a block we want and a block we observe, with the observed block held at a , the result is normal with

$$\bar{\mu} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(a - \mu_2), \quad \bar{\Sigma} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Here $N = 2$ with one variable in each block. We observe x_1 and want x_2 , so x_2 takes the role of the first block and x_1 the second:

$$\text{Var}[x_2 \mid x_1] = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

The factor we are asked for is the ratio of that to the unconditional variance:

$$\frac{\text{Var}[x_2 | x_1]}{\text{Var}[x_2]} = 1 - \frac{\Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}}{\Sigma_{22}}$$

The sample covariance matrix is

$$\Sigma = \begin{bmatrix} 1.0999 & 1.6969 \\ 1.6969 & 4.1047 \end{bmatrix}$$

so $\Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12} = 1.6969^2 / 1.0999 = 2.6180$, and

$$\text{Var}[x_2 | x_1] = 4.1047 - 2.6180 = 1.4867$$

The variance falls to $1.4867 / 4.1047 = 0.3622$ of its unconditional value, so about 64% of the variance is removed. In standard deviation terms the drop is from 2.0260 to 1.2193, a factor of 0.6018.

Each block is a scalar here, so the ratio reduces to $1 - \rho^2$ with $\rho = 0.7986$, and the standard deviation factor to $\sqrt{1 - \rho^2} = 0.6018$. That simplification is particular to the two variable case. The block form is what generalizes.

The term that settles part (b) is $\Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$ in $\bar{\Sigma}$. It is built from the covariance blocks alone, and the observed value a appears nowhere in it. Under the multivariate normal the conditional variance is therefore a constant, the same whichever x_1 turns up. Only $\bar{\mu}$ moves with a .

4.2 Parts (c) through (e)

Taking the same partition, the conditional mean is

$$E[x_2 | x_1] = \mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (x_1 - \mu_1)$$

The coefficient on x_1 is $\Sigma_{21} \Sigma_{11}^{-1} = 1.6969 / 1.0999 = 1.5428$. In regression terms it is the slope from regressing x_2 on x_1 , which is the connection Week 2 draws between conditioning a normal vector and fitting a regression. The conditional variance is the constant from part (a), 1.4867, so the band has half width $1.96\sqrt{1.4867} = 2.39$ at every value of x_1 .

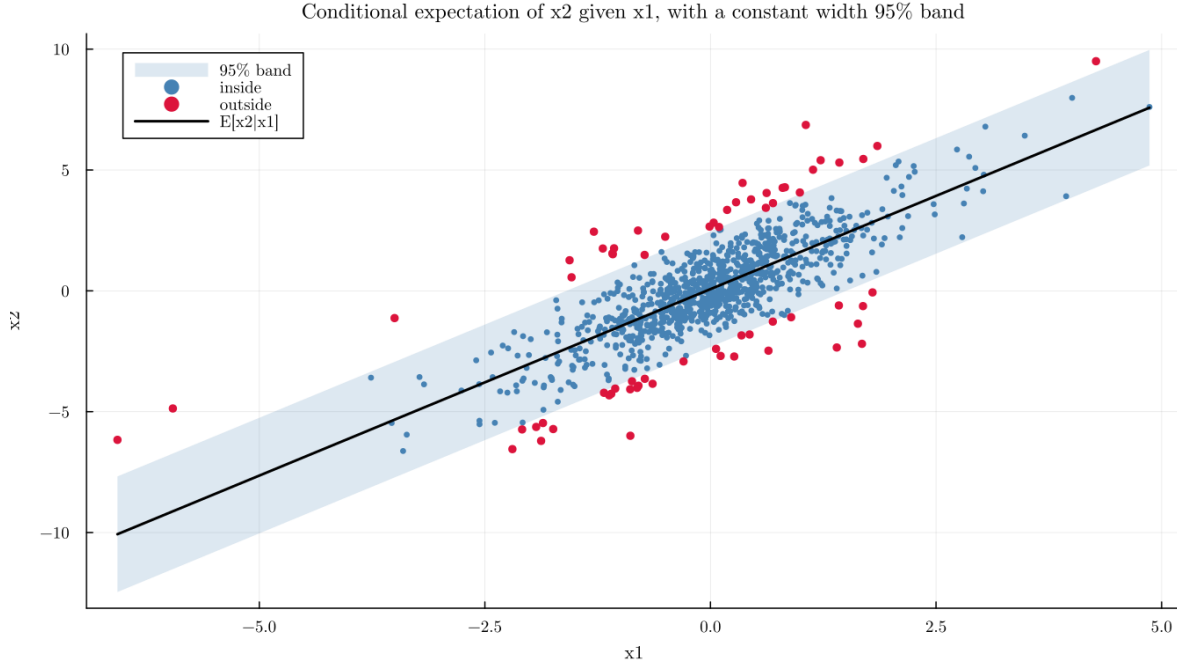


Figure 3: The conditional expectation with a constant width 95% band. The misses concentrate in the shoulder rather than at the extremes, and the band is visibly too wide in the far tails

Overall coverage of the 95% band is **93.5%**, close enough to 95% to look like a pass. The misses in Figure 3 are not spread evenly, which is what the split below makes precise.

Splitting on $|x_1 - \mu_1|$ measured in standard deviations:

Table 5: Coverage of the constant width band, by distance from the mean

Bucket	n	Coverage
Inside 1 sd	759	95.7%
1 to 2 sd	190	85.8%
Beyond 2 sd	51	90.2%

4.3 Part (f)

The coverage is not flat. It sits at or above nominal through the middle and falls to 85.8% in the shoulder. The overall 93.5% hides the pattern, because 76% of the sample sits in the bucket where the band works.

The data is elliptical but not normal. For an elliptical distribution the conditional mean stays linear in x_1 , so the regression line from part (c) is right and stays right. The conditional variance does not stay constant: it grows with how far x_1 sits from its mean. A band of fixed width is therefore too wide near the middle and too narrow out in the shoulder, which is the pattern in Table 5.

The assumption that failed is normality, and specifically the constant conditional variance that normality delivers. The linear conditional mean survives, because it follows from ellipticity alone.

The “beyond 2 sd” bucket is not the worst, at 90.2% against 85.8%. With only 51 observations in it, that difference sits inside sampling noise and we would not read much into the ordering of those two buckets.

5 Identifying an AR or MA Order

5.1 Parts (a) and (b)

Table 6: ACF and PACF of `problem5.csv`

Lag	1	2	3	4	5	6
ACF	0.462	-0.025	-0.166	-0.110	-0.084	-0.041
PACF	0.462	-0.304	-0.017	-0.018	-0.084	0.014

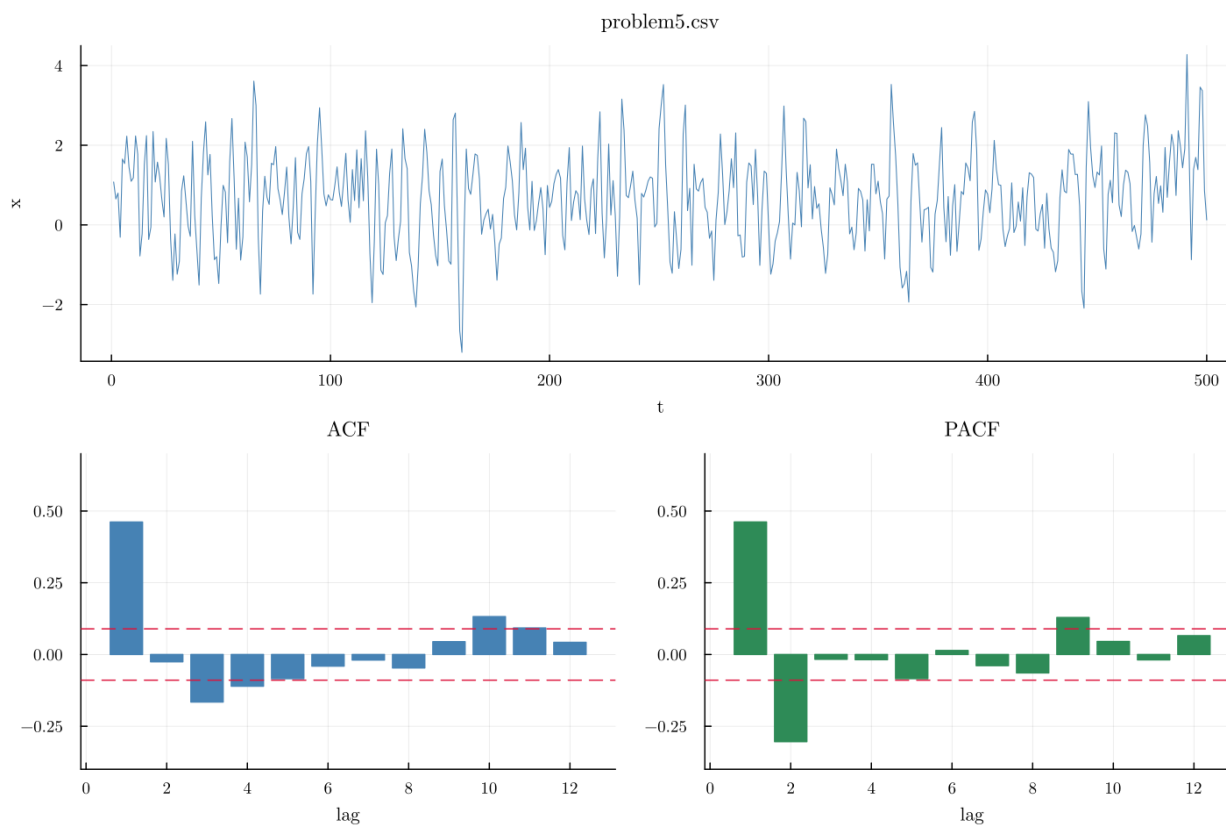


Figure 4: The series, its ACF, and its PACF. The dashed lines are $\pm 2/\sqrt{n}$

The significance band is $2/\sqrt{n} = 0.089$, drawn on Figure 4.

In Table 6 the PACF cuts off after lag 2. Lags 1 and 2, at 0.462 and -0.304 , sit far outside the band, and lags 3 through 8 are all well inside it.

The ACF does not cut off. It runs 0.462, -0.025 , -0.166 , -0.110 , -0.084 , -0.041 across the first six lags, tailing toward zero with no point after which it stays there. The magnitude does not fall monotonically, since lag 2 is small and lag 3 is larger again, and the sign changes along the way. Neither matters for the identification. The rule asks whether the function cuts off or decays, and this one decays.

Week 2 gives the rule directly. An ACF that decays paired with a PACF that cuts off after lag p is an $\text{AR}(p)$, and here the cutoff is at lag 2. The reading is **AR(2)**.

Two features of Figure 4 do not change that. The PACF carries an isolated spike at lag 9, 0.129 against a band of 0.089, and the ACF sits marginally outside at lags 10 and 11. Twelve lags tested at the 5% level produce about one exceedance by chance, so an isolated spike with nothing around it is noise. The ACF values at lags 3 and 4 are a different case: they sit inside the decay described above rather than standing alone, which is what the identification rule is reading.

5.2 Parts (c) and (d)

Table 7: AICc across the six candidate models

Model	AICc	Fitted parameters
AR(1)	1412.67	$\phi = 0.462$
AR(2)	1366.56	$\phi = (0.602, -0.304)$
AR(3)	1368.47	$\phi = (0.597, -0.294, -0.017)$
MA(1)	1383.33	$\theta = 0.530$
MA(2)	1375.33	$\theta = (0.633, 0.163)$
MA(3)	1368.15	$\theta = (0.594, 0.065, -0.156)$

Table 7 selects AR(2), matching what we read off the plots.

The MA models climb toward the AR(2) fit as their order rises, and MA(3) at 1368.15 nearly reaches it. That is expected, since a finite order AR inverts to an infinite order MA and a long enough MA approximates it. AICc still prefers the two parameter model.

5.3 Part (e)

AR(3) fits marginally better in likelihood terms and is charged for the third parameter. The estimated $\phi_3 = -0.017$ is small and sits inside the PACF's significance band, so the extra parameter buys almost nothing. AICc charges about 2 units per parameter, the log likelihood improved by less than that, and the larger model loses by 1.91.

R^2 would not have made this choice. R^2 cannot fall when a parameter is added, so it weakly prefers AR(3) over AR(2) and would keep preferring larger models indefinitely. It measures fit to the sample in hand and charges nothing for complexity. That is why Week 2 introduces the information criteria: they estimate out of sample performance, and R^2 does not try to.

Adjusted R^2 does carry a penalty and would likely also pick AR(2) here, though its penalty is not derived from a prediction argument.